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Measurement of self-diffusion in thin samples using a novel one-sided NMR magnet

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ABSTRACT

A novel one-sided NMR magnet has been validated for accurate measurements of diffusion in thin samples. Results are presented from the measurement of self-diffusion in several solvents, including the room temperature ionic liquid 1-butyl-3-methyl imidazolium bis(trifluoromethane)sulfonimide (BMIM-TFSI). We also present results from diffusion measurements of BMIM-TFSI confined in nanoporous anodized aluminum oxide (AAO) membranes (thickness 100 μm), a promising electrolyte/membrane combination for portable energy devices. A reduction in the diffusion coefficient, compared to the bulk, has been observed in BMIM-TFSI confined in narrow pores (55 nm). The possible application of this instrument to *in situ* measurements of portable energy devices is discussed.

1. Introduction

Research and development of novel and improved portable energy devices – e.g. batteries, electrochemical double-layer capacitors (EDLCs; often referred to as "supercapacitors") and fuel cells - forms a substantial part of the large wave of activity by scientists focused on improving all aspects of the so-called global energy mix. As pressures increase on existing fossil fuel reserves, it is expected that portable energy devices will become more important as a key ingredient in delivering efficient, reliable and sustainable energy, especially in the significant area of automotive transport. A crucial component of this scientific and technological endeavor is the identification and development of advanced materials with high ion conductivity for application as electrolytes [1].

Nuclear Magnetic Resonance (NMR) is a valuable technique for studying and characterizing these emerging materials *ex situ*, and through methodological advancements over the past decade, *in situ* NMR studies of portable energy devices have been shown to be not only feasible, but valuable in revealing important information of the workings of the devices, non-invasively, and in real-time [2–7].

Despite the impressive progress that has been made to date in the application of NMR to the study of portable energy devices, the geometry of a typical NMR spectrometer used for these kinds of studies places constraints on the shape of the device that can be studied, and severely limits access to the device during measurements. Also, the components of the devices (electrodes, separating membranes) are often porous, which leads to problems with local magnetic field gradients due to differences in magnetic susceptibility between liquid and the host matrix, an effect that is more significant in stronger magnetic fields.

In situ NMR studies of portable energy devices would benefit from the availability of unconventional instruments that relax some of these constraints. In this context, low-field (i.e. $B < 1.0$ T) unilateral NMR instruments may offer several advantages for such studies. In particular, the one-sided geometry of these instruments opens up possibilities of combining NMR measurements with other measurement techniques (e.g. optical measurement), and allows devices of different form factors to be studied. Also, working at low-field minimizes the local magnetic-field inhomogeneity caused by susceptibility effects, which in turn results in longer effective T_2 values, and the possibility to measure smaller diffusion coefficients.

The magnetic field profiles of unilateral magnets for NMR are, in most cases, largely inhomogeneous, therefore the acquisition of high-resolution spectra is especially challenging, although it has been demonstrated that chemical shift information can be recovered via z-rotation pulses if the RF field profile is matched with the static field [8,9]. However, on the other hand, spatially-resolved measurements of relaxation and spin-density, and measurements of self-diffusion coefficients D can be performed routinely with one-sided systems [10–14].

In this paper we: (i) validate a novel one-sided NMR magnet [15] by demonstrating its capability to accurately measure self-diffusion in liquids; (ii) report measurements of self-diffusion in

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thin porous samples that are technologically relevant to portable energy devices; in particular we present self-diffusion measurements of an ionic liquid (IL) electrolyte [16,17] confined in porous alumina membranes. The alumina membranes are non-reactive and electrically insulating and as such they can be used as separators between the electrodes of portable energy devices. The membranes have hexagonally packed, cylindrical pores, with monodisperse diameters. The pores are aligned, running straight between the surfaces of the membrane, and are open at both ends. The fabrication by anodic etching gives fine control over the pore diameter and the inter-pore spacing [18].

2. Experimental

The one-sided NMR magnet consists of an assembly of NdFeB permanent magnets. The magnetic field is designed to produce a well-controlled magnetic field gradient perpendicular to the apparatus' surface (Fig. 1), whilst retaining a field profile in which the magnitude of the magnetic field strength $|B|$ is very flat in extended planes parallel with the apparatus' surface. This particular geometry of the apparatus and the magnetic field profile makes it ideally suited to NMR measurements of self-diffusion in thin, extended samples.

Compared to existing one-sided magnets that have been reported in the literature, this magnet can be placed in the category of systems that have a large region of interest, in which the magnetic field has a constant gradient (e.g. Perlo et al. [19], Garcia-Naranjo et al. [14]). This genre of device is suitable for acquiring one-dimensional projections of macroscopic objects without the need for physically displacing the object or sensor. On the other hand, devices such as the NMR-MOUSE [12] or GARFIELD [20] purposefully sacrifice the size of the sensitive volume in order to prioritize portability and/or instrument resolution.

The instrument has an 'optimized region' which has a thickness of 8 mm and an outer diameter of 10 mm; the center of the optimized region is located 20 mm from the surface of the magnet, and at this point the magnetic field strength is 0.33 Tesla, corresponding to a proton Larmor frequency of 14.08 MHz. In the transverse plane at the center of the optimized region, field variation is less than 100 ppm, which is currently unrivalled by similar magnets of this type. The magnetic field gradient B_z is equal to 3.3 Tm^{-1} in this region (measured via a Hall probe, and Cartesian mapping with an NMR micro-coil [15]). The instrument's spatial resolution – as determined by the point spread function of a 1D projection of a flat sample [15] – is 20 μm, which is competitive with the current state-of-the-art [12].

The coupling of the low magnetic field with a strong gradient gives very favorable conditions for measuring D in slowly diffusing liquids, confined in porous media. These measurements may be performed in the stray fields of superconducting magnets [10], however this approach is less suitable for large samples, due to the strong curvature of the magnetic field.

The NMR probe comprised a rectangular solenoid coil, which was wound to accommodate 10 mm wide glass cells used for holding samples. The thickness of the sample space of the glass cells was 1 mm. Utilizing the solenoid coil afforded the greatest sensitivity for these samples, however it should be noted that this instrument is also capable of single-sided measurements of larger samples when equipped with a surface coil [15].

Diffusion was measured using a Hahn-echo pulse sequence, appended with a CPMG train; the echoes were summed in order to increase the signal-to-noise ratio, as is the common procedure for NMR measurements in inhomogeneous fields [10,11]. Attenuation of the amplitude of the spin-echo in a constant gradient field is given by

$$I(2\tau) = I_0 \exp\left(\frac{-2\tau}{T_2} - \frac{2}{3}\gamma^2 D g^2 \tau^2\right) \tag{1}$$

where I is the amplitude of the spin echo, τ is the echo time, g is the magnetic field gradient, and γ is the gyromagnetic ratio. T_2 was measured for each sample by acquiring a train of Hahn-echoes, using a CPMG sequence with an echo time of 0.5 ms, i.e. short enough to avoid attenuation due to diffusion, and long enough to observe a significant decrease in the echo amplitude. Under these conditions, the measured T_2 value is an effective value that is influenced by the variation of the flip-angle in the inhomogeneous B_0 field, but effects due to diffusion may be neglected. The decay curves of the echo amplitude versus echo time were fitted to a single-exponential decay function.

The ionic liquid 1-butyl-3-methyl imidazolium bis(trifluoromethane)sulfonimide (BMIM-TFSI) was purchased from Solvionic (France), and used as received. Nanoporous anodized aluminum oxide (AAO) membranes were purchased from SmartMembranes GmbH (Germany) and Synkera Technologies Inc. (USA), and used as received. The dimensions of the membranes were 10 mm × 10 mm × 0.1 mm.

Membranes were loaded with BMIM-TFSI by submerging them in the ionic liquid for several days. Small Angle Neutron Scattering (SANS) measurements [21] on these samples have shown that this is an effective method for loading the pores with ILs. For fully loaded membranes, assuming that confinement of the IL does not affect its density, the mass of absorbed liquid is on the order of 1 mg. The membranes were pressed between two glass plates in order to position them in the NMR coil for measurement. When in place in the instrument, the long axes of the pores in the membrane are aligned with the magnetic-field gradient, therefore the

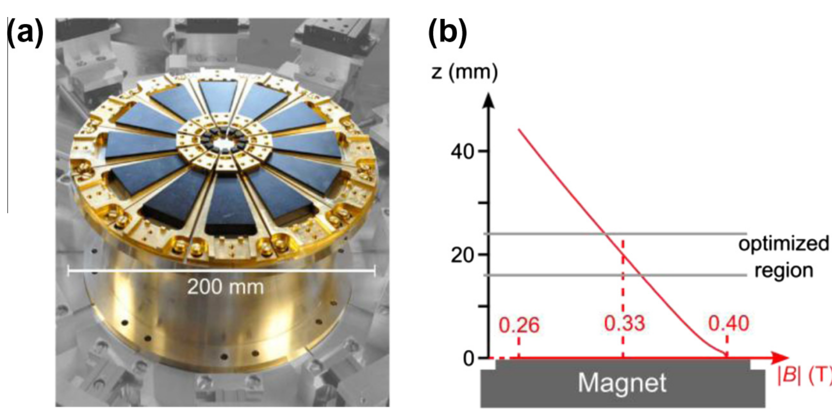

Fig. 1. (a) Photograph of the one-sided NMR magnet. (b) Schematic diagram of the variation in the magnitude of the magnetic field strength $|B|$ at the axial centre of the apparatus, as a function of distance from the surface. The field profile is designed so that the planes of constant $|B|$ are most flat in the optimised region.

measurements are not sensitive to transverse diffusion in the pores.

The experimental time for measuring the diffusion coefficient of bulk liquids was around 8 min (not including approximately 1 min for measurement of T_2). For the membrane-confined liquid, measurement of the proton self-diffusion coefficient and T_2 required acquisition times of around 24 h each.

3. Results

3.1. Instrument qualification

In Fig. 2a the results of ^{1}H self-diffusion measurements are plotted for three common laboratory solvents. The effective T_2 values measured for these samples were in the range 3–8 ms – compared to diffusion times of up to 1.6 ms – therefore transverse relaxation during the diffusion measurement was non-negligible. In such cases, according to Eq. (1), D can be determined by plotting $\ln\left(\frac{I}{I_0}+2\tau/T_2\right)$ as function of τ^2 and fitting the data to a straight line. The resultant self-diffusion coefficients are presented in Table 1, along with accepted values from the literature, with which there is good agreement.

In order to probe the uniformity of the optimized region, ^{1}H self-diffusion was measured in water at three positions in the optimized region (Fig. 2b), i.e. $\Delta z = -1, 0, 3$ mm, where Δz denotes vertical displacement of the center of the sample, relative to the center of the optimized region (N.B. the combined thickness of the sample stage and coil prevented measurements in the region $\Delta z < -1$ mm). The values of $D(^1\text{H})$ was measured to be $(2.3 \pm 0.1) \times 10^{-5}$ cm^2 s^{-1} at each position, an excellent agreement that is also clear from inspection of Fig. 2b.

These results demonstrate that the value of the average magnetic field gradient in the optimized region of the instrument, determined by field-mapping, is reliable, and that the instrument is capable of accurate measurements of self-diffusion.

3.2. Self-diffusion in bulk and confined ionic liquid

Due to the presence of fluorine, and the absence of protons, in the chemical structure of TFSI (see inset of Fig. 3), the self-diffusion coefficient of the cation and anion can be determined separately, by observing the ^{1}H and ^{19}F nuclei, respectively. For samples of the given thickness of 1 mm, the signals from each nucleus are separated sufficiently in order to allow independent measurements (i.e. the distance between the ^{1}H and ^{19}F sensitive planes is around 6 mm, which is much larger than the sample thickness).

Having knowledge of both D_{cation} and D_{cation} allows transference numbers to be estimated for each ionic species, which are amongst the properties that characterize a substance for application as an electrolyte [17]. Measurements of D_{cation} and D_{cation} are presented in Fig. 3 for BMIM-TFSI. The corresponding self-diffusion coefficients are presented in Table 2, alongside previously reported values [22], with which there is very good agreement.

Data from ^{1}H self-diffusion measurements of BMIM-TFSI confined in two AAO membranes with pore diameters of 55 nm and

Table 1
Measured self-diffusion coefficients of some common laboratory solvents compared with accepted values from literature.

	$D(T = 25\,°\text{C})/10^{-5}$ cm^2 s^{-1}		T_2 (ms)
	Measured	Literature[a]	
Acetone	4.4 ± 0.1	4.57	3.4
Water	2.3 ± 0.1	2.30	4.7
Ethanol	1.1 ± 01	1.08	8.4

[a] Reported in the 2012 Bruker Almanac.

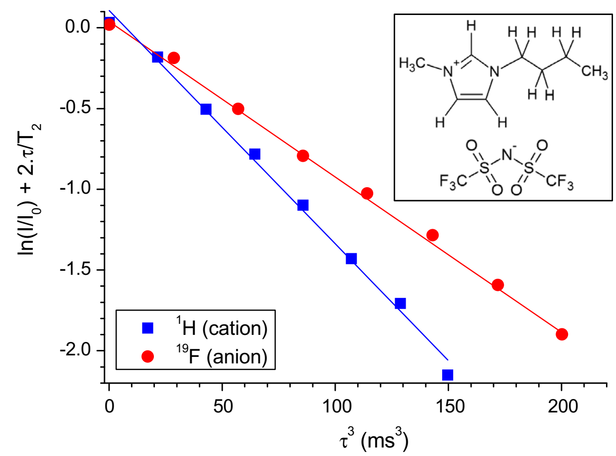

Fig. 3. Measurements of the cation and anion self-diffusion coefficients in the ionic liquid BMIM-TFSI (bulk measurement). Inset: chemical structure of BMIM-TFSI.

Table 2
Measured self-diffusion coefficients of cation and anion in BMIM-TFSI compared with values reported in the literature.

	$D(T = 25\,°\text{C})/10^{-7}$ cm^2 s^{-1}		T_2 (ms)
	Measured	Literature[b]	
D_{cation} (BMIM)	2.8 ± 0.1	2.8	28
D_{cation} (TFSI)	2.1 ± 0.1	2.2	49

[b] Tokuda et. al. [22].

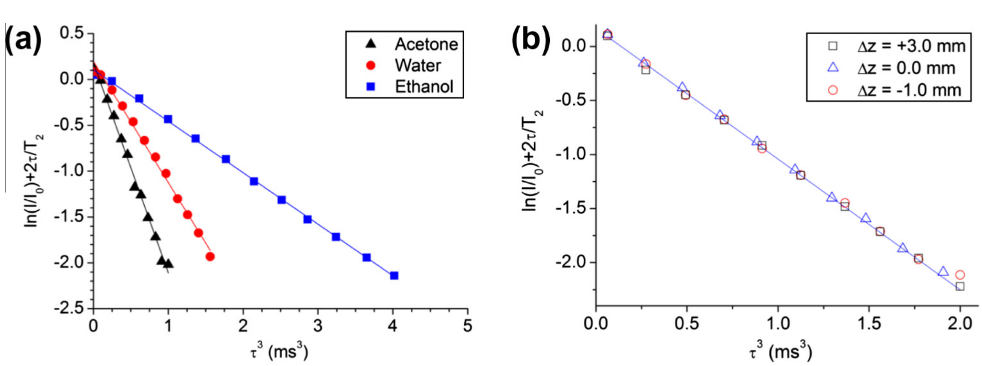

Fig. 2. Instrument qualification: (a) ^{1}H self-diffusion measurements for common laboratory solvents acetone, ethanol, and distilled water; (b) ^{1}H self-diffusion measurement in water, at three different positions in the optimized region. Δz denotes vertical displacement of the center of the sample, relative to the center of the optimized region. Solid line corresponds to fit of data measured at $\Delta z = 0$ mm.

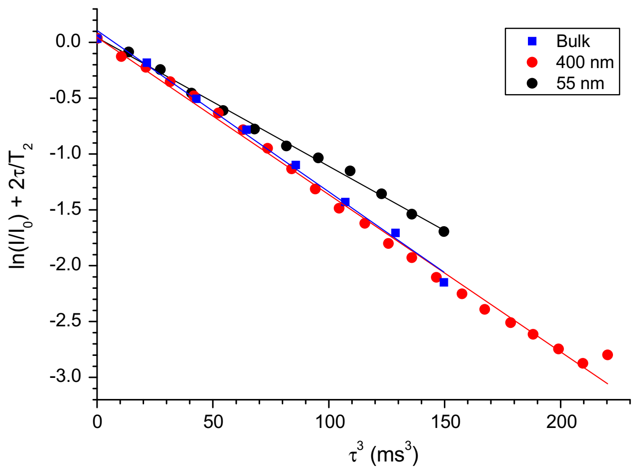

Fig. 4. Measurements of the cation self-diffusion coefficient of BMIM-TFSI, confined in AAO membranes with pore diameters of 55 nm and 400 nm. For comparison, data from bulk ionic liquid are also plotted.

400 nm are presented in Fig. 4, and compared with bulk. As with the bulk measurements, the diffusion data from the confined IL display a single-component. The diffusion coefficient of the IL cation in the 400 nm pores was measured to be $(2.7 \pm 0.1) \times 10^{-7}$ $cm^2\ s^{-1}$, which is in agreement with the value measured in bulk, within experimental error. A reduced diffusion coefficient of $(2.7 \pm 0.1) \times 10^{-7}\ cm^2\ s^{-1}$ was measured in the 55 nm pores. The proton T_2 for BMIM-TFSI confined in 55 nm and 400 nm pores was 27 ± 2 ms and 23 ± 1 ms, respectively. The possible origins of the difference in the diffusion coefficients are explored in Section 4.

4. Discussion

All measured values of the self-diffusion coefficients in bulk liquids have been found to be in good agreement with expected values; this confirms that the apparatus is well-characterized, with a carefully determined magnetic field profile.

Measurements of the cation self-diffusion coefficient in membrane-confined ionic liquids were possible; conversely, equivalent measurements for the anion were not possible, due to the low number of ^{19}F nuclei present in the sample. The ^{19}F NMR signal was indeed observable and diffusion data with reasonable signal-to-noise ratio were acquired, however correction of these data for T_2 relaxation was not possible, as the sensitivity of the spectrometer precluded an accurate measurement of T_2 in a practicable time. Confinement of the liquid in the pores and the interactions with the pore surface causes T_2 to be modified, and therefore it is not appropriate to assume the value of the bulk T_2 for the confined liquid.

The similarity between the measured values of $D(^1H)$ for the bulk IL and the IL confined in pores of 400 nm diameter suggests that confinement at this scale does not significantly modify the diffusion properties of the IL, although a 20% reduction was observed for IL confined in 55 nm pores. Given the value of the bulk diffusion coefficient, the estimated diffusion length for BMIM during a diffusion time of 5 ms (i.e. approximately the longest diffusion time used) is 530 nm, which well exceeds the pore diameter; the diffusion is therefore expected to be anisotropic. In such cases an apparent diffusion coefficient (ADC), which is smaller than the free diffusion coefficient, can be observed if the pores are not aligned parallel to the magnetic field gradient. For cylindrical pores aligned at an angle θ with respect to the magnetic-field gradient, the ADC is given by

$$\mathrm{ADC} = D_{\parallel}\cos^2\theta + D_{\perp}\sin^2\theta \quad (2)$$

Where $D_{||}$ and $D_{\perp}$ are the longitudinal and transverse diffusion coefficients, respectively. If $D_{||}$ is assumed to be equal to the self-diffusion coefficient measured in bulk, and $D_{\perp}$ is estimated to be much smaller than $D_{||}$, a 20% reduction in the diffusion coefficient, as has been observed, would require the sample to be tilted by 28° with respect to the magnet surface. This orientation is beyond the range of the sample holder therefore sample-tilting can be excluded as the cause of observed reduction in the observed diffusion coefficient. In principle, orientation-dependent measurements of the apparent diffusion coefficient are possible using this instrument, although measurements of this nature have yet to be undertaken.

Interactions between the ionic liquid and the surface of the AAO pores can be expected to play a role in the transport properties of the confined liquid. Fast exchange between surface-bound molecules and the "bulk" population (i.e. liquid not in contact with the surface) would result in an average diffusion coefficient that is smaller than the bulk value, an effect that would increase with decreasing pore diameter. That said, if a relatively immobile monolayer of ionic liquid is assumed to exist at the pore surface, it is estimated that the relative difference of the surface-to-volume ratio of the 55 nm pores compared to the 400 nm pores, would result in only a 5% reduction in the diffusion coefficient. Thus this simple model is inadequate in accounting for the observed 20% reduction of the diffusion coefficient of the ionic liquid confined in the narrowest pores. The actual cause of the reduction may be due to a combination of effects, including surface interactions, and possible departure of the pore geometry from the assumed ideal cylindrical geometry.

5. Conclusion

The ability of the one-sided NMR magnet to make reliable measurements of self-diffusion coefficients has been demonstrated for bulk samples, and also for very thin porous samples in which the liquid mass is no more than a few milligrams. Based on this work it is expected that the one-sided geometry of the system will allow self-diffusion measurements and one-dimensional imaging in suitable portable energy devices. For example, measurements in fuel cells and ionic-liquid based EDLCs would be possible due to the abundance of protons in these devices; however it remains to be determined if the sensitivity of this instrument would allow *in situ* observations of the technologically important ^7Li nucleus.

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